09-KKT-exc

Generic

1. Reminding the linearization formula for a scalar differentiable funcion $f\colon \mathbb{R}^n \to \mathbb{R}$ near $\mathbf{x}$: 4439eb

$$f(\mathbf{x}+\mathbf{h}) - f(\mathbf{x}) \simeq \nabla f(\mathbf{x}) \circ \mathbf{h} = (\nabla f)^T \mathbf{h} = (\nabla^T f)\mathbf{h} = [(\partial_{x_1}, \dots, \partial_{x_n})f]\,\mathbf{h}.$$

   (a) Show that the linearization of a differentiable field (or, why not, change of variables) $\mathbf{g}\colon \mathbb{R}^n \to \mathbb{R}^n$:

$$\mathbf{g}(\mathbf{x}+\mathbf{h}) - \mathbf{g}(x) \simeq \begin{bmatrix} \partial_{x_1} g_1 & \dots & \partial_{x_n} g_1 \\ & \vdots & \\ \partial_{x_1} g_n & \dots & \partial_{x_n} g_n \end{bmatrix} \begin{bmatrix} h_1 \\ \vdots \\ h_n \end{bmatrix} = \begin{bmatrix} -\nabla^T g_1 - \\ \vdots \\ -\nabla^T g_n - \end{bmatrix} \begin{bmatrix} h_1 \\ \vdots \\ h_n \end{bmatrix} = \underbrace{\left(\frac{d\mathbf{g}}{d\mathbf{x}}\right)}_{J} \mathbf{h}, \quad J\colon \text{ the Jacobian.}$$

   Check that the Jacobian of $\mathbf{g}(\mathbf{x}) = A\mathbf{x}$ is, of course, $J = \left(\frac{d\mathbf{g}}{d\mathbf{x}}\right) = A$.

   (b) Let now $\mathbf{f}, \mathbf{g}\colon \mathbb{R}^n \to \mathbb{R}^n$ be two vector fields. Show that the gradient of their dot product

$$\nabla(\mathbf{f}^T\mathbf{g}) = \nabla(\mathbf{g}^T\mathbf{f}) = \left(\frac{d\mathbf{f}}{d\mathbf{x}}\right)^T \mathbf{g} + \left(\frac{d\mathbf{g}}{d\mathbf{x}}\right)^T \mathbf{f} = J_{\mathbf{f}}^T \mathbf{g} + J_{\mathbf{g}}^T \mathbf{f}.$$

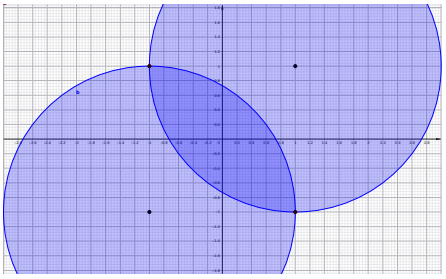
   (c) In particular, show that gradient of a quadratic form $\nabla(\mathbf{x}^T A\mathbf{x}) = (A + A^T)\mathbf{x}$.

   (d) Conclude by showing that the Hessian of the quadratic form $\mathbf{x}^T A\mathbf{x}$ is given by

$$\begin{bmatrix} \partial^2_{x_1} & \cdots & \partial^2_{x_1 x_n} \\ \vdots & \ddots & \vdots \\ \partial^2_{x_n x_1} & \cdots & \partial^2_{x_n} \end{bmatrix} (\mathbf{x}^T A\mathbf{x}) = \nabla\nabla^T(\mathbf{x}^T A\mathbf{x}) = \nabla[\nabla(\mathbf{x}^T A\mathbf{x})]^T = A + A^T.$$

KKT 2D

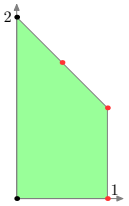
1. Investigate whether $(-1, 1)$ is a KKT point for the NLP c4add1



$$\max\ f(\mathbf{x}) = x_1^2 + x_1 + 2x_2$$
$$\text{s.t.} \begin{cases} \dfrac{(x_1-1)^2}{4} + \dfrac{(x_2-1)^2}{4} \le 1 \\ \dfrac{(x_1+1)^2}{4} + \dfrac{(x_2+1)^2}{4} \le 1. \end{cases}$$

   Can you claim that a KKT point is a global max?

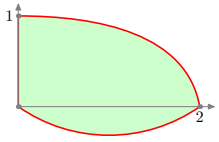
2. Consider the non-linear optimization problem: 3b7988



$$\begin{cases} \min & f(\mathbf{x}) = x_1^2 - 2x_1x_2 + 2x_2^2 - 2x_1 - 4x_2 \\ \text{s.t.} & x_1 + x_2 \le 2, \quad x_1 \le 1, \quad x_{1,2} \ge 0. \end{cases}$$

   Is any of $(1, 0)$, $(\frac{1}{2}, \frac{3}{2})$ or $(1, 1)$: (a) a KKT point? (b) a global min point?

3. Consider the non-linear problem: 76cc65

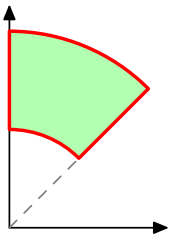


$$\max\ f(\mathbf{x}) = x_1^2 - 3x_1x_2 + 2x_2^2$$
$$\text{s.t. } \mathbf{x} \in \Omega\colon \begin{cases} x_1^2 + 4x_2^2 \le 4 \\ x_2 \ge x_1^2 - 2x_1 \end{cases}$$

   (a) Is the problem convex (i.e., Ω is a convex set and f, a concave function)? (b) Is (2,0) a KKT point?

4. Consider the non-linear constrained problem f40d18

$$\max\ f(\mathbf{x}) = x_1^2 + x_1 + x_2^2,$$
$$\mathbf{x} \in \Omega\colon \begin{cases} 1 \le x_1^2 + x_2^2 \le 8 \\ 0 \le x_1 \le x_2. \end{cases}$$



(a) Determine whether the problem is convex.

(b) Investigate if $\mathbf{x} = (2, 2)$ is a KKT point for the problem above.

5. The LP below is solved by the Simplex Method with a Final Tableau as follows: 7bad83

$$\max \quad z = \begin{bmatrix} 6 & 10 & 12 \end{bmatrix} \mathbf{x},\ \mathbf{x} \ge \mathbf{0},$$
$$\text{s.t.} \quad \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \le \begin{bmatrix} 20 \\ 70 \end{bmatrix}$$

$\perp$	x_1	x_2	x_3	s_1	s_2	rhs
z	8	0	0	4	2	220
x_2	11	1	0	4	-1	10
x_3	-8	0	1	-3	1	10

(a) Reformulate the LP as a normal max problem $\{\max f(\mathbf{x}),\ \text{s.t. } \mathbf{g} \le \mathbf{0}\}$.

(b) Write down the corresponding Lagrangian $\mathcal{L}(\mathbf{x}, \boldsymbol{\mu})$.

(c) Check explicitly that the KKT conditions are met at the optimal point $\mathbf{x}^*$.

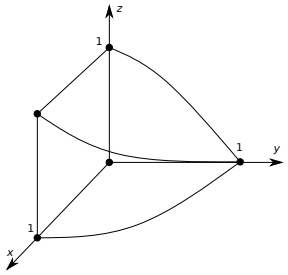
6. Rewrite the non-linear problem to the right in normal form and sketch its feasible domain. Does any of the points (0,2), (2,0), (1,0) satisfy the KKT conditions?
Can you claim that any of them is a global min? (An affirmative answer requires a solid motivation).

$$\min f(\mathbf{x}) = x_1^2 + 2x_1x_2 + 2x_2^2$$
$$\text{s.t.} \begin{cases} x_1^2 + 2x_2 \le 4 \\ 2x_1 + x_2 \ge 2 \\ x_2 \ge 0 \end{cases}$$

KKT 3D

1. Consider the non-linear optimization problem 7adb4e

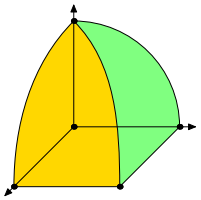
$$\max\ f(x, y, z) = (x^2 - 2z)e^y,$$
$$(x, y, z) \in \Omega\colon \begin{cases} x^2 + y^2 \le 1, \\ x \ge 0,\ y \ge 0, \\ 0 \le z \le 1 - y^2. \end{cases}$$



Which constraints are active at the point $A = (1, 0, 1)$? Is A a KKT-point?

2. Consider the NLP: dbc1e3

$$\min\ f(\mathbf{x}) = (x_1 - x_3)^2 + x_2^2 - 4(x_1 + x_2),$$
$$\text{s.t. } \mathbf{x} \in \Omega\colon \begin{cases} x_3 \le 1 - x_1^2 \\ x_3 \le 1 - x_2^2 \\ x_1 \ge 0,\ x_2 \ge 0,\ x_3 \ge 0. \end{cases}$$



Rewrite it in standard form and motivate whether the point $A = (1, 1, 0)$ is

(a) a KKT-point? (b) a global minimum point for $f_{|\Omega}$?